Serhii Hirus

AI Automation & Backend Engineer · Remote, full-time

Portfolio · LinkedIn · GitHub

SUMMARY

Automation engineer working at the intersection of Python, LLM APIs (Claude, OpenAI, Vertex AI) and n8n. I design fault-tolerant systems: autonomous AI agents, RAG knowledge bases, CRM synchronization and media pipelines for product and engineering teams. Open to AI Automation Engineer, Python / AI Solutions Engineer and LLM Integration Developer roles.

SKILLS

- **AI agents and LLM pipelines:** LangGraph, tool calling, structured outputs (Pydantic), Vertex AI with context caching, retries and model fallbacks
- **RAG and knowledge bases:** Qdrant, pgvector, LangChain, hybrid search with reranking
- **Automation and integrations:** n8n (self-hosted and cloud), Make, webhooks, REST APIs, HubSpot, Pipedrive, Telegram, Slack
- **Backend and data:** Python (FastAPI, Asyncio, Pydantic), SQL, PostgreSQL, Playwright, JavaScript / Node.js
- **Media and speech:** Whisper, FFmpeg
- **DevOps:** Docker, Linux, Git / GitHub, CI/CD, Google Cloud

SELECTED PROJECTS

Automated Content Creator

Python · Claude API · Whisper · FFmpeg · n8n · PostgreSQL · Docker · Google Cloud

- Built an autonomous pipeline that turns videos and articles into posts and subtitled clips for 6 platforms (LinkedIn, Telegram, X, Threads, Medium, Discord).
- Telegram bot as the control interface, Python microservices (parsing, Whisper, FFmpeg rendering) and a PostgreSQL scheduler.
- Claude API prompts and structured outputs adapt tone and style to each platform.
- **Result:** replaces 15+ hours of manual work per week (sourcing, clipping, subtitling, writing, posting) with one-click publishing or scheduled autoposting.

Quant Trading Lab (hirus.systems)

Python · Vertex AI · VectorBT · Optuna · FastAPI · Neon PostgreSQL · React 19

- Built a platform that turns trading hypotheses written in natural language into algorithmic strategies, with automated out-of-sample backtesting and overfitting protection.
- Tuned Vertex AI with a 35.4k-token context cache for the VectorBT API; validated generated Python with static AST analysis (blocking eval/exec) before execution; added an automatic self-healing loop.
- Implemented an async FastAPI backend with vectorized simulation in VectorBT and Bayesian optimization in Optuna.
- **Result:** strategy generation, syntax verification and backtest in under 3 seconds (4x faster with context caching); 100% of generated code passes AST validation; full cycle from hypothesis to walk-forward analysis.

EDUCATION

Courses and certifications

- Google: Introduction to Generative AI, 2026.
- Coursera: Generative AI with Large Language Models, 2026.

LANGUAGES

English — B1 · Ukrainian — native · Russian — native